

THE BOEING COMPANY — QUALITY ENGINEERING

STANDARD OPERATING REPORT | DAY 26: SUPPLIER QUALITY MANAGEMENT

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Standard:	AS9100D / ISO 9001:2015 / ISO 19011	Author:	Jagriti Singh Quality Engineer
Subject:	Tier-1 Supplier Non-Conformance & Precision Fastener Risk Mitigation		

1. EXECUTIVE SUMMARY

This quality engineering report establishes the supplier non-conformance containment and risk mitigation protocol for commercial aerospace operations at Boeing. By targeting critical precision fasteners and sub-assembly hardware, this process integrates ISO 19011 supplier quality audits, real-time Statistical Process Control (SPC), and automated assembly-line Poka Yoke to ensure zero-defect component delivery, eliminate structural assembly bottlenecks, and enforce strict aerospace compliance.

2. OPERATIONAL PROBLEM STATEMENT

- Target Components:** Precision Structural Fasteners & Mechanical Assembly Hardware.
- Core Issue:** Dimensional variance, material tolerance drift, and thread specification non-conformance from Tier-1 suppliers.
- Manufacturing Impact:** Non-conforming fasteners result in shear test failures, thread stripping, assembly line stoppages, and costly manual rework loops during fuselage and wing section joint integration.

3. BENCHMARKING & QUALITY ARCHITECTURE

- Aerospace Benchmarking Target:** Calibrated against AS9100D standard requiring process capability index $C_{pk} \geq 2.00$ (Six Sigma capability, ensuring defect rate threshold below 3.4 Defects Per Million Opportunities / PPM).
- ISO 19011 Supplier Audits:** On-site audit protocols verifying vendor tool calibration, raw material traceability, and shop-floor SOP execution prior to dispatch authorization.
- Supplier SPC ($\bar{X}$ & R Charts):** Real-time process capability monitoring at supplier plants to identify process drift before parts leave the facility.
- Torque-Limit Poka Yoke:** Electronic and mechanical torque-sensing assembly tools deployed on the shop floor to eliminate over-tightening and thread stripping at point-of-use.

4. ENGINEERING IMPACT & TARGET METRICS

Metric Indicator	Baseline (Current)	Benchmark Target	Applied TQM Mechanism
Incoming Rejection Rate	4.2%	< 0.00034%	ISO 19011 Audits & Supplier SPC ($C_{pk} \geq 2.00$)
First-Pass Yield (FPY)	91.5%	99.99%	Torque-Limit Poka Yoke Integration
Assembly Rework Time	High Loop Delay	35% Reduction	8D Containment & Standardized Fastening SOPs

5. CORRECTIVE ACTION & AUTHORIZATION SIGN-OFF

Root Cause Protocol:	Closed via 8D Problem Solving Methodology	Implementation:	APPROVED
Authorized By:	Jagriti Singh (Quality Engineer)	Scope:	Tier-1 Aerospace Fastener Suppliers

Academic & Professional Disclaimer: This document was prepared by Jagriti Singh as part of the #100DaysOfTQM educational initiative. The methodologies, quality architecture, and benchmarking models presented herein are derived from standard industrial quality engineering principles (AS9100D / Six Sigma / ISO 19011) and serve purely as an educational case study.